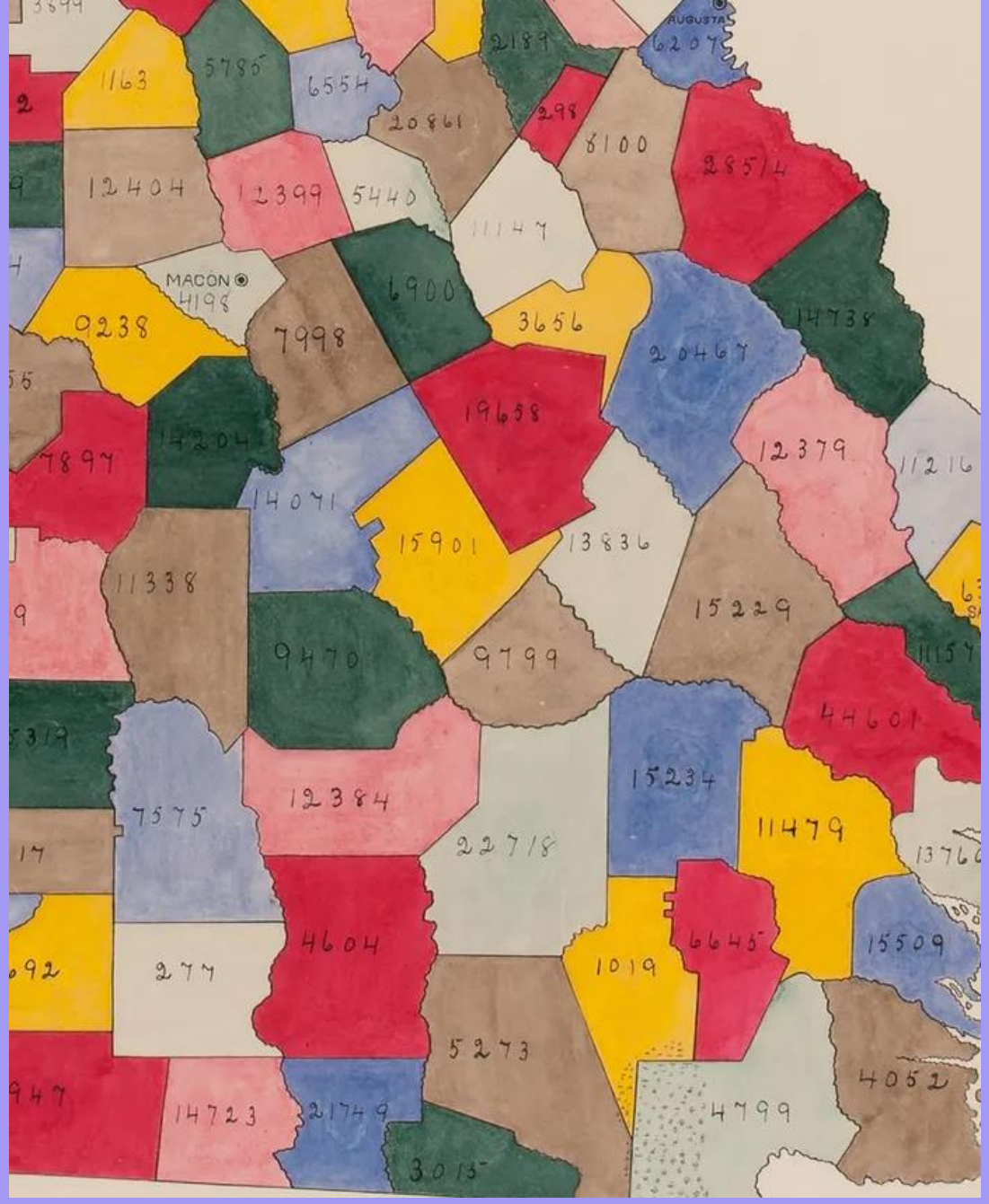
Inclusive Mapping Workshop:
Creating a Design Brief

27th October | City St George's University of London

Part of the ESRC's Digital Good Network Project:
Can Digital Goods be Neutral? *Evaluating OpenStreetMap's Equity Through Participatory Data Visualisations*

UNIVERSITY OF WARWICK | CITY ST GEORGE'S UNIVERSITY OF LONDON | ESRC Digital Good NETWORK



Thank you!

Because you've decided to spend several hours with us.

Because your participation will be key at providing insights for our research project.

On the other hand, we expect **that by the end of today you will**

- Have a renewed understanding of the relationship between maps, inequalities and neutrality,
- Have an opportunity to reflect and discuss about the role of maps and data visualisations to promote/hamper equity and inclusion
- Have learnt specialised tools
- Have met nice and interesting people
- Have fun!

About us



Carlos Cámara-Menoyo (PI)
Senior Research Software Engineer
Centre for Interdisciplinary Methodologies



Timothy Monteath (Co-I)
Assistant Professor
Centre for Interdisciplinary Methodologies



Mirela Reljan-Delaney
Lecturer in Computer Science
City St George's, University of London

Geochicas:
Selene Yang,
Alejandra Canclini,
Silvia Ribera-Alfaro

Advisory board:
Jorge León *(Universidad de Zaragoza)*,
Rachel Palmen *(Universitat Oberta de Catalunya)*,
Meg Davis *(University of Warwick)*
Selene Yang *(Stanford Center on Philanthropy and Civil Society)*

About the project

Can Digital Goods be Neutral? Evaluating OpenStreetMap's equity through participatory data visualisation.

- One of the 9 funded projects by the **ESRC Digital Good Network** (Research Fund 2024)
- Our research enquires about how **neutrality** is operationalised to favour or hinder **equity** in a particular digital good: **OpenStreetMap**.

What do maps have to do with equity/gender/ethnicity...? Aren't they objective? Can you give **examples**?

That's what we are asked a lot when we present our project.

And that's why we are here today!

Inclusive cartographies workshop series

Producing the brief
18th October, 24th October (online)

Producing the cartographic style
27th October







Online evaluation

Aims and agenda

Produce a **cartographic style** for an **inclusive cartography** from a given **design brief**

12:00 – 12:20	**Introductions** (Carlos Cámara)
12:20 – 12:35	Talk: **Maps and inequalities** (Mirela Reljan-Delaney)
12:35 – 12:50	Task: **Group formation**
12:50 – 13:40	Tools: **OpenStreetMap and QGIS** (Timothy Monteath, Carlos Cámara)
13:40 – 14:30	Lunch
14:30 – 14:30	Task: **Producing the cartographic style** (Groupwork)
16:25 – 17:00	**Discussion and evaluation** (Timothy Monteath)

Housekeeping

We will speak about potentially sensitive topics: gender, race, sexual orientation…

- Mediated through the use of fictional personas
- No need to disclose any personal information
- Be respectful and considerate with each other

We will be collecting the following information:

- Photos* and recordings
- Workshop’s materials (Personas, Maps…)
- Anonymous survey
- Notes (see Chatham House rules)

* If you want to opt out, please let us know.

> “ anyone is free to use information from the discussion/workshop, but is not allowed to reveal who made any particular comment.
>
> Chatham House Rules

Sign the **consent form**, please!

We need it to use the outputs in the research project



01
Maps and inequalities

Mirela Reljan-Delaney

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02

Tools: OSM and QGIS

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Group formation

Form groups to create the cartographic style (10’)

1. Read all the design briefs
2. Stay around the brief you would like to implement today.
3. Wait for others to join your table/area.
4. When 3 or more people are gathered around, you have formed a group!
    - Introduce yourselves to the colleagues
5. Assign roles:
    - QGIS
    - Documenter
    - Optional:
        - OSM Wiki / taginfo
        - Designer (e.g. icons, colours…)



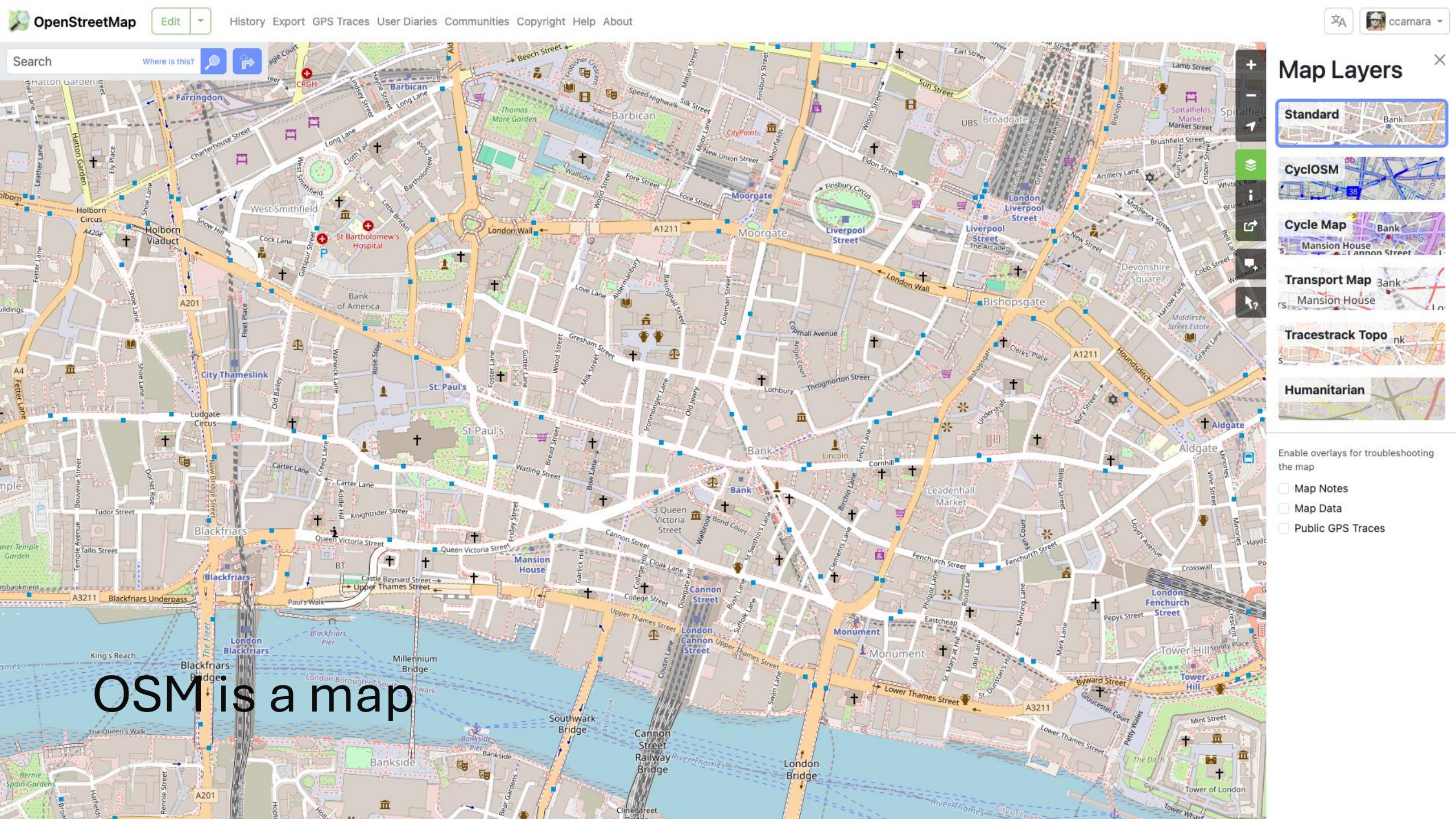
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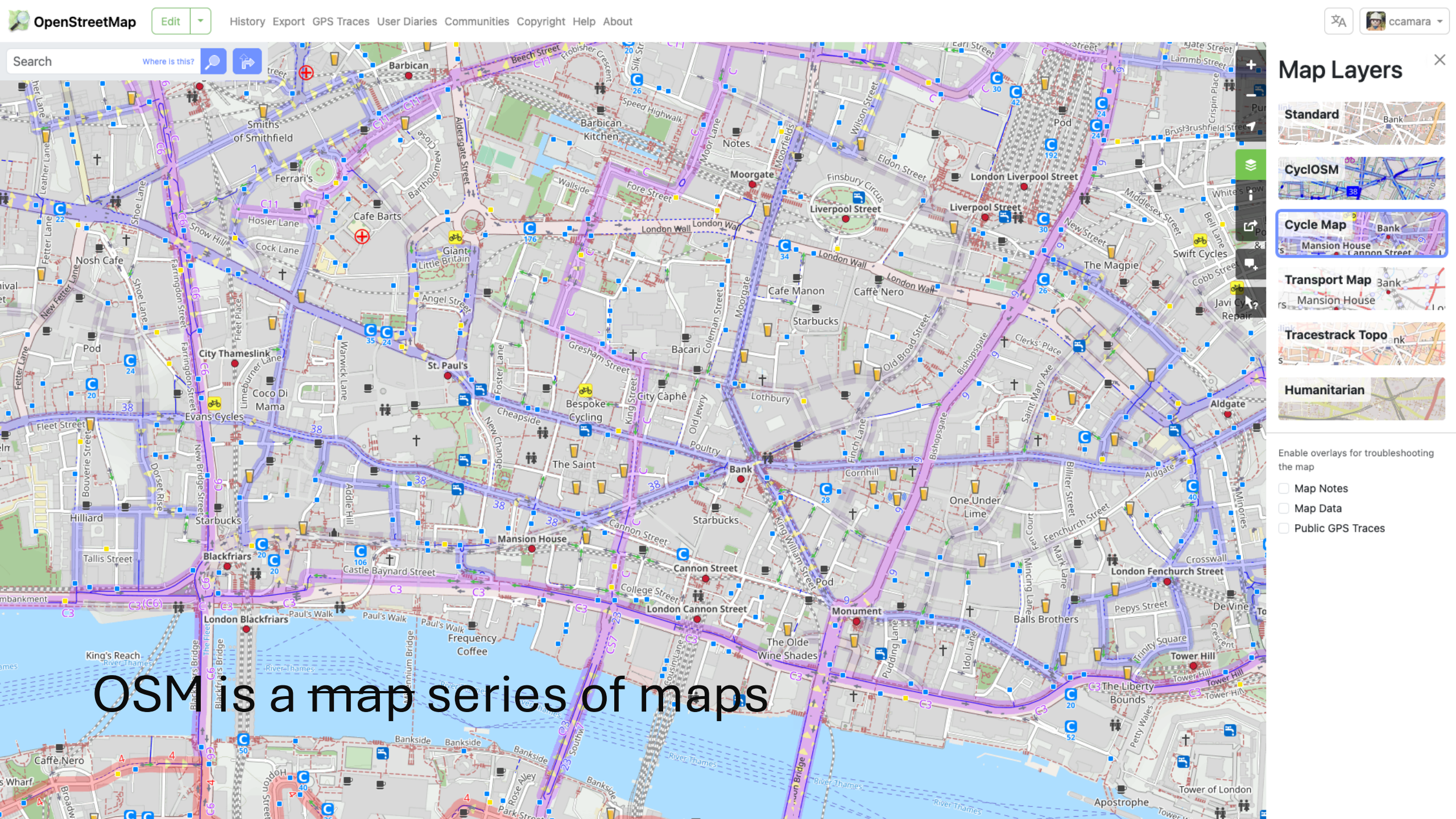


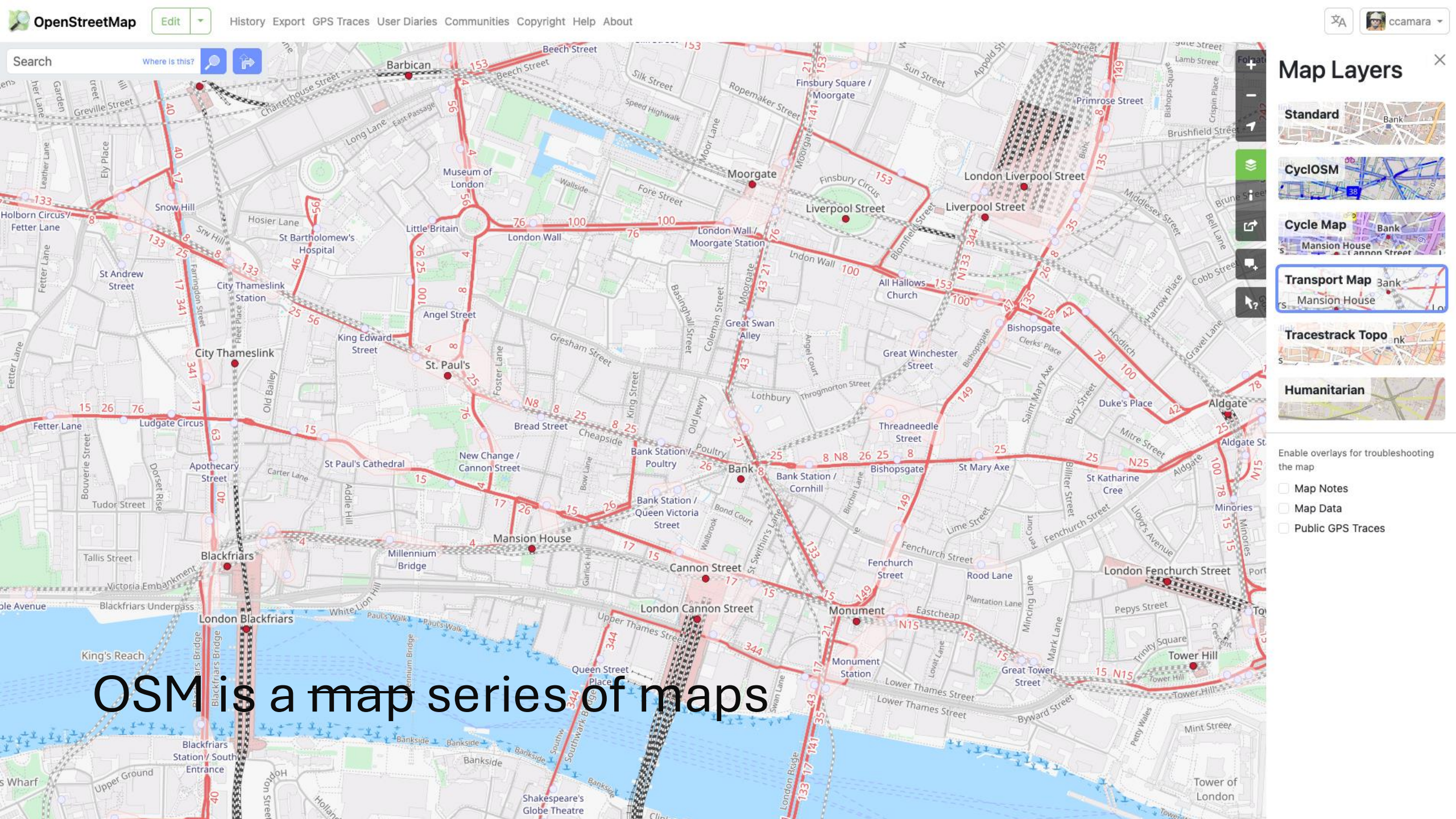
“The Wikipedia of the maps”

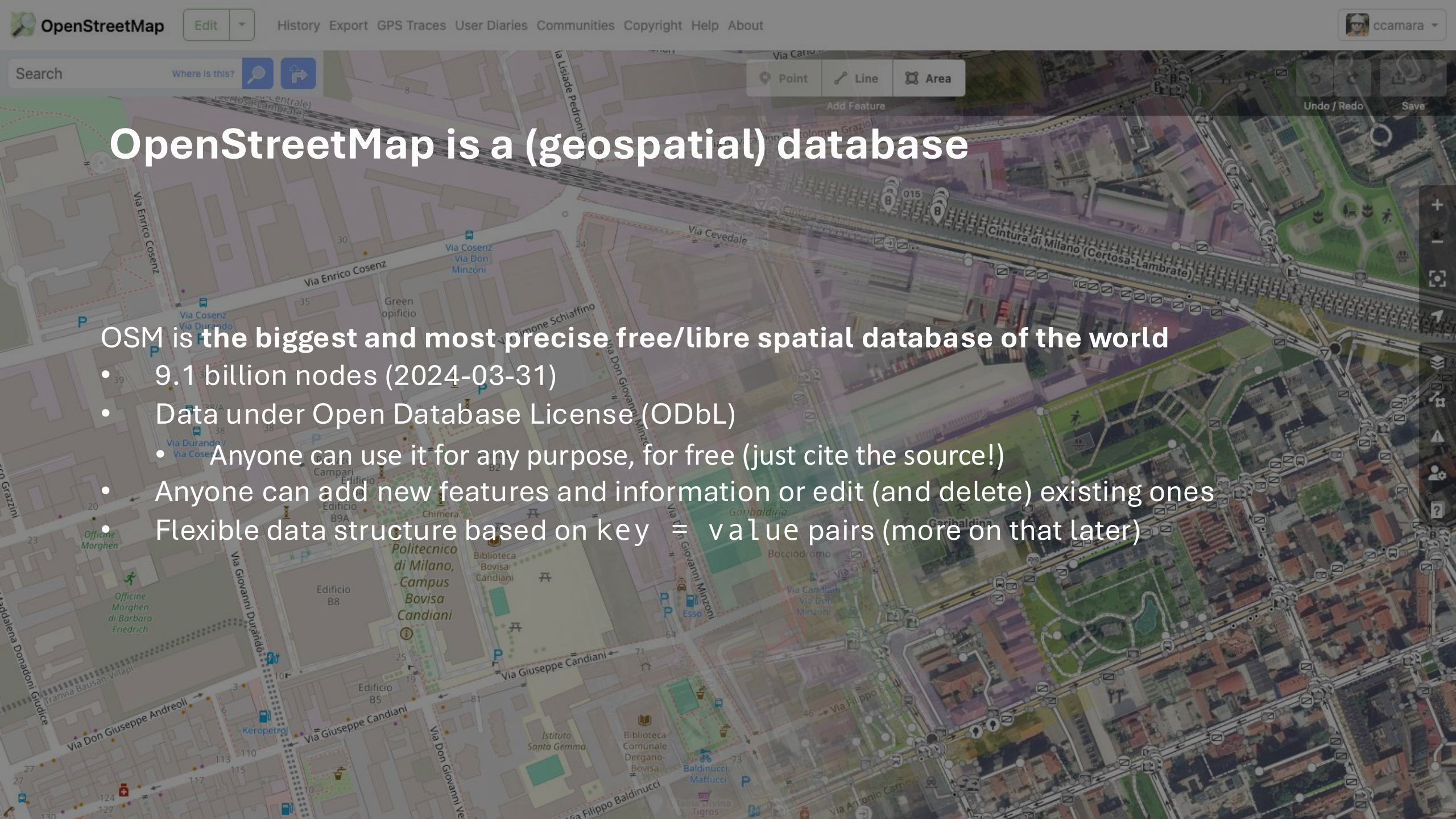
OpenStreetMap

https://openstreetmap.org


OpenStreetMap
Edit
History
Export
GPS Traces
User Diaries
Communities
Copyright
Help
About
ccamara
Search
Where is this?
Map Layers
Standard
CyclOSM
Cycle Map
Transport Map
Tracestrack Topo
Humanitarian
Enable overlays for troubleshooting the map
Map Notes
Map Data
Public GPS Traces
OSM is a map


OpenStreetMap
Edit
History Export GPS Traces User Diaries Communities Copyright Help About
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Humanitarian
Enable overlays for troubleshooting the map
Map Notes
Map Data
Public GPS Traces
OSM is a ~~map~~ series of maps

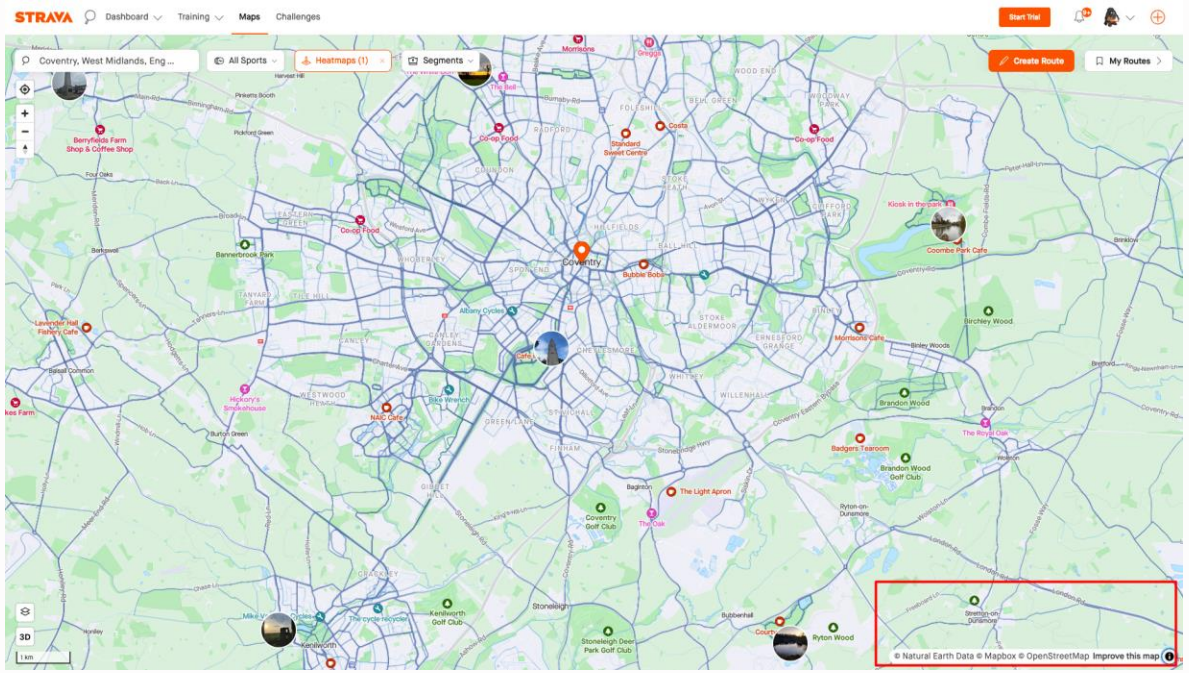

OpenStreetMap
Edit
History Export GPS Traces User Diaries Communities Copyright Help About
ccamara
Search
Where is this?
Map Layers
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Transport Map
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Map Notes
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OSM is a ~~map~~ series of maps

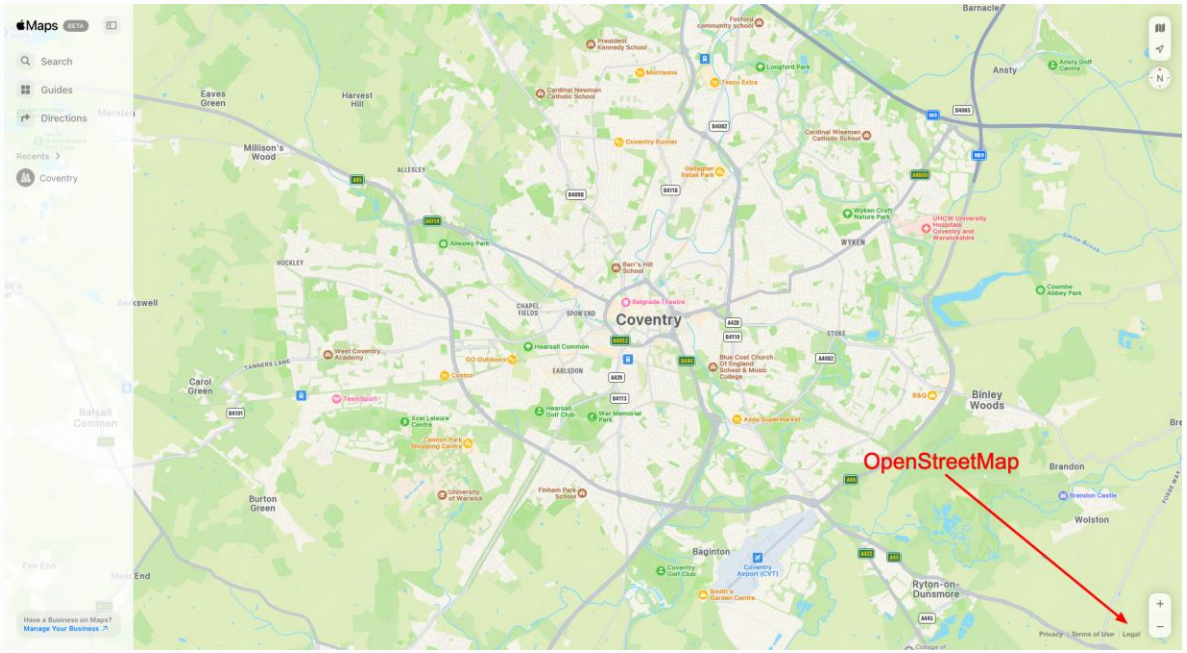


OpenStreetMap is a (geospatial) database

OSM is **the biggest and most precise free/libre spatial database of the world**

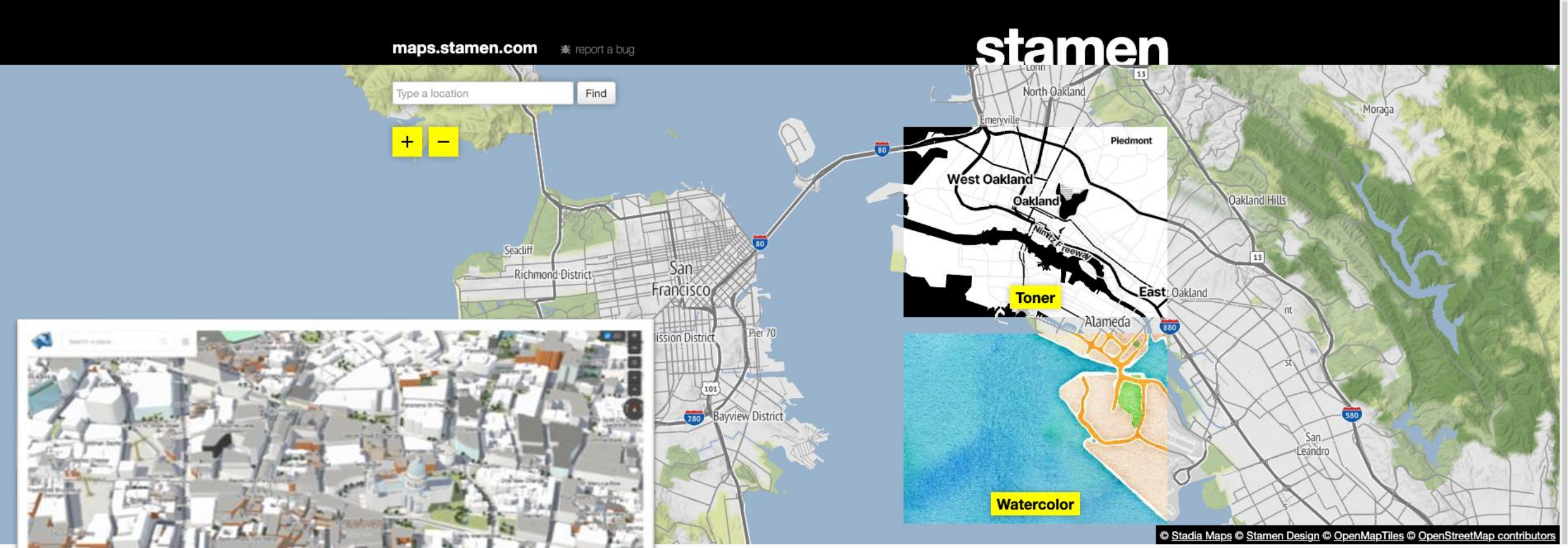
- 9.1 billion nodes (2024-03-31)
- Data under Open Database License (ODbL)
  - Anyone can use it for any purpose, for free (just cite the source!)
- Anyone can add new features and information or edit (and delete) existing ones
- Flexible data structure based on `key = value` pairs (more on that later)





Its data, contributed by 10 million of volunteers, complements official data sources, and **populates thousands of tools and services**.

Major Sites: Amazon, Apple, Baidu Maps, Facebook, Microsoft, Wikipedia and Wikimedia
Transport: Air France, Alaska Airlines, Deutsche Bahn, Grab, SNCF (French rail agency), Uber
Geodata Software and Services: CARTO, Digital Globe, ESRI, Garmin, Mapbox, Telenav
Government: Agence Française de Développement , Government of Brazil, Government of Italy: President's Office, Police Scotland, US National Park Service, US State Department, USAID, Peace Corps....



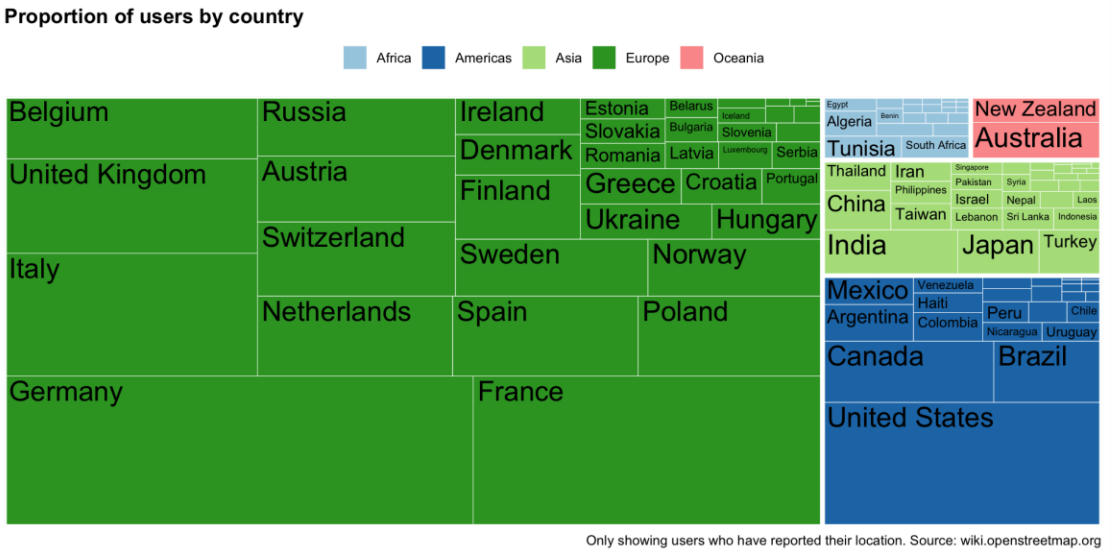


Some of these companies create cartographic styles with OSM data for others to use: Stamen, Carto, Mapbox, Stadia, Mapzen…

OSM is a community (of communities)

More than **10 million users worldwide**, mainly volunteers (but not always)

- All users have **equal level of permissions** to:
  - Add new features to the map
  - Edit existing features to add more detail or remove outdated details
  - Remove
  - Decide what or how to organise the information in the dataset (i.e. what is included and what is not)
- Users may be part of local communities
- The OSM Foundation oversees the OSM project and has several WG



Like Wikipedia, **OSM is based on** principles of **openness** and **neutrality.**

“**OpenStreetMap maps world as it exists**, and includes mapping borders and countries according to actual current situation and **not a preferred or ideal situation**” (OSM Wiki)

What can be mapped?



English Ccamara Talk Preferences Watchlist Contributions Log out

Page Discussion Read Edit Edit source More Search OpenStreetM

Main Page
The map
Map Features
Contributors
Help
Blogs
Shop
Donations
Wiki discussion
Recent changes

Scope

OpenStreetMap welcomes diverse contributions from our diverse community. Defining what falls within OSM remit is, therefore, challenging. As a rule of thumb, anything that ticks all the following conditions would be welcome:

- It is verifiable,
- It is mappable, within the planet Earth,
- It is not private or copyrighted information,
- It is compatible with the OpenStreetMap License,
- It is added manually (see the community guidelines for automated edits),
- It is not listed under Limitations.

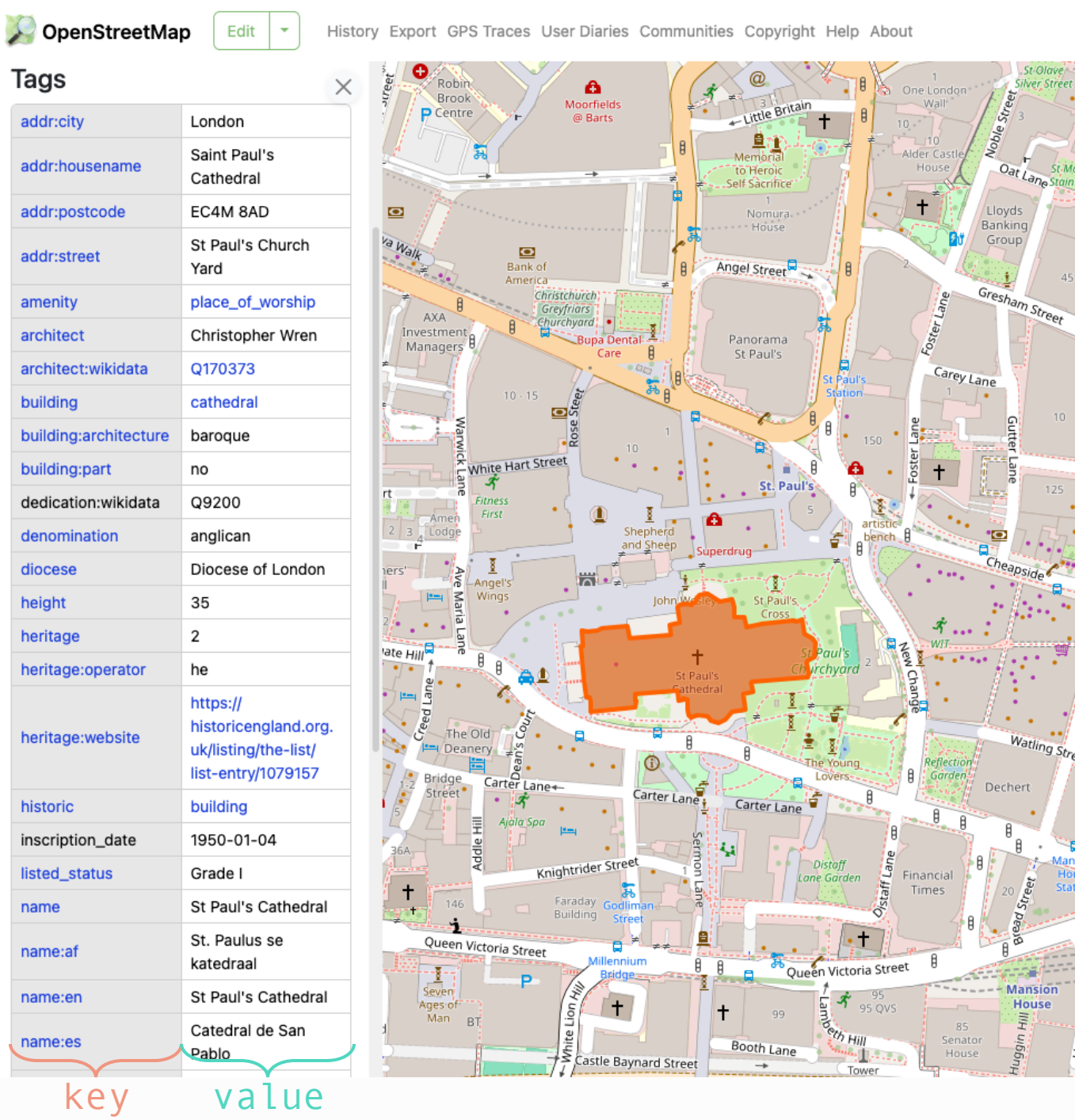
Could these types of data be included in OSM?

- Population’s data (e.g. census)
- Historical data
- Descriptions about a shop
  - menus,
  - Stairs
  - LGTBIQ+ friendliness
  - ratings
- Data from governments, city councils, police…

How are things mapped?

Understanding OSMs data structure

- In OSM, a **feature** is a physical element in the world subject to be mapped (street, building, sign, tree, bench…).
- Each OSM feature is described in two dimensions:
  - Spatial: through their coordinates (latitude and longitude) and element type (point, line, polygon, relationships)
  - **Meaning:** through tags (free format text fields)
- Each **tag** is defined by a pair of `key = value`
  - Keys can be assimilated to categories or topics
  - Value provide detail for the key-specified feature
- Not all keys are displayed on a map (rendered)
  - There's way more information that remains hidden (usually, for good reasons)
  - This information can be queried!



Querying OSM data

Using Overpass Turbo: https://overpass-turbo.eu/

Run | Share | Export | Wizard | Style | Save | Load | Settings | Help | overpass turbo | Map | Data

```
/*
This has been generated by the overpass-turbo wizard.
The original search was:
""Drinking Water""
*/
[out:json][timeout:25];
// gather results
node["amenity"="drinking_water"]({{bbox}});
// print results
out geom;
```



This data can be exported (downloaded) and used to create our own spatial dataset (foundation for a map)

We will be querying Overpass API using a QGIS

How do we know what to query in Overpass Turbo?

In short: OSM Wiki and Taginfo

OSM wiki: collaborative documentation

https://wiki.openstreetmap.org

The wiki describes the consensus reached by OSM community.

Keys and Tags are described in OSM wiki. The wiki explains how to use them:

- Description/meaning
- Applicable elements
- List of accepted values
- Whether it is rendered or not
- ...

We can use the search feature to find how to tag certain features.

The wiki has links to two other important services:

- Overpass Turbo
- Taginfo



English Ccamara Talk Preferences Watchlist Contributions Log out

Page Discussion Read Edit Edit source View history More Search OpenStreetMap

Main Page
The map
Map Features
Contributors
Help
Blogs
Shop
Donations
Wiki discussion
Recent changes

Tools

What links here
Related changes
Upload file
Special pages
Printable version
Permanent link
Page information
Cite this page
Data item

Tag:amenity=drinking_water

(Redirected from Drinking Water)

Tag:amenity=drinking water · **In other languages** Help

čeština Deutsch **English** español français Nederlands polski português Ελληνικά русский українська 한국어 中文（繁體） 日本語

Other languages...

Drinking water is a place where humans can obtain potable water for consumption. Typically, the water is used for drinking. This includes **drinking fountains** or **bubblers**, water taps, water wells, springs and water points. Generally, this feature provides small quantities of water.

Where drinking water is provided in large quantities, use `amenity=water_point` or `waterway=water_point`; i.e., for refilling tanks in motorhomes, boats, etc.

Contents [hide]

1 How to map
  1.1 Useful combinations
  1.2 Nonworking features
2 Alternative tagging
3 Photo examples
4 Proposal history
5 External links
6 Common/possible tagging mistakes
7 See also

amenity = drinking_water v·d·e



Description

A drinking water source which provides potable water for consumption.

Rendering in OSM Carto

Group: amenities

Used on these elements

Implies

- drinking_water=yes

How to map [edit | edit source]

Add `amenity=drinking_water` to an existing element or create a new one.

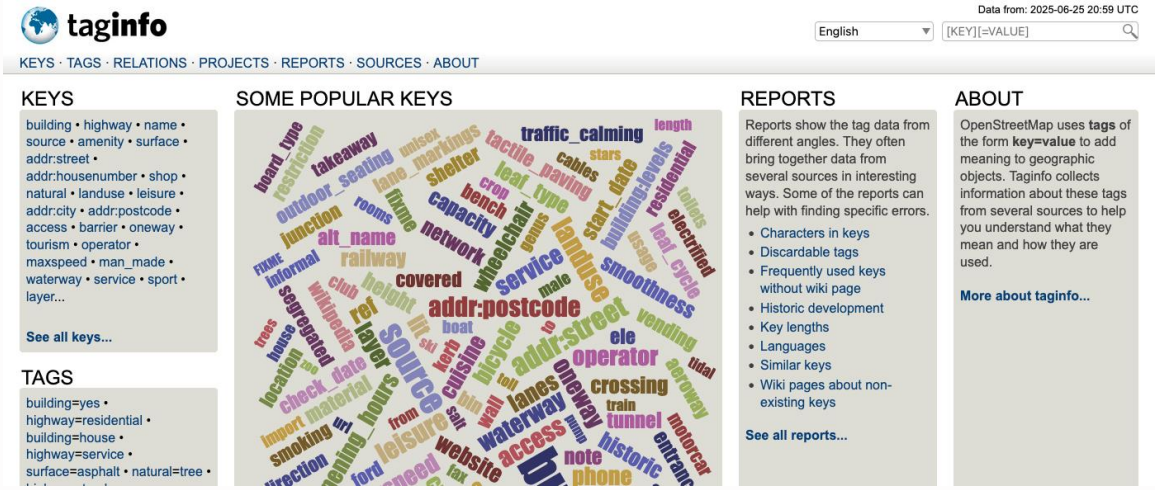
Taginfo

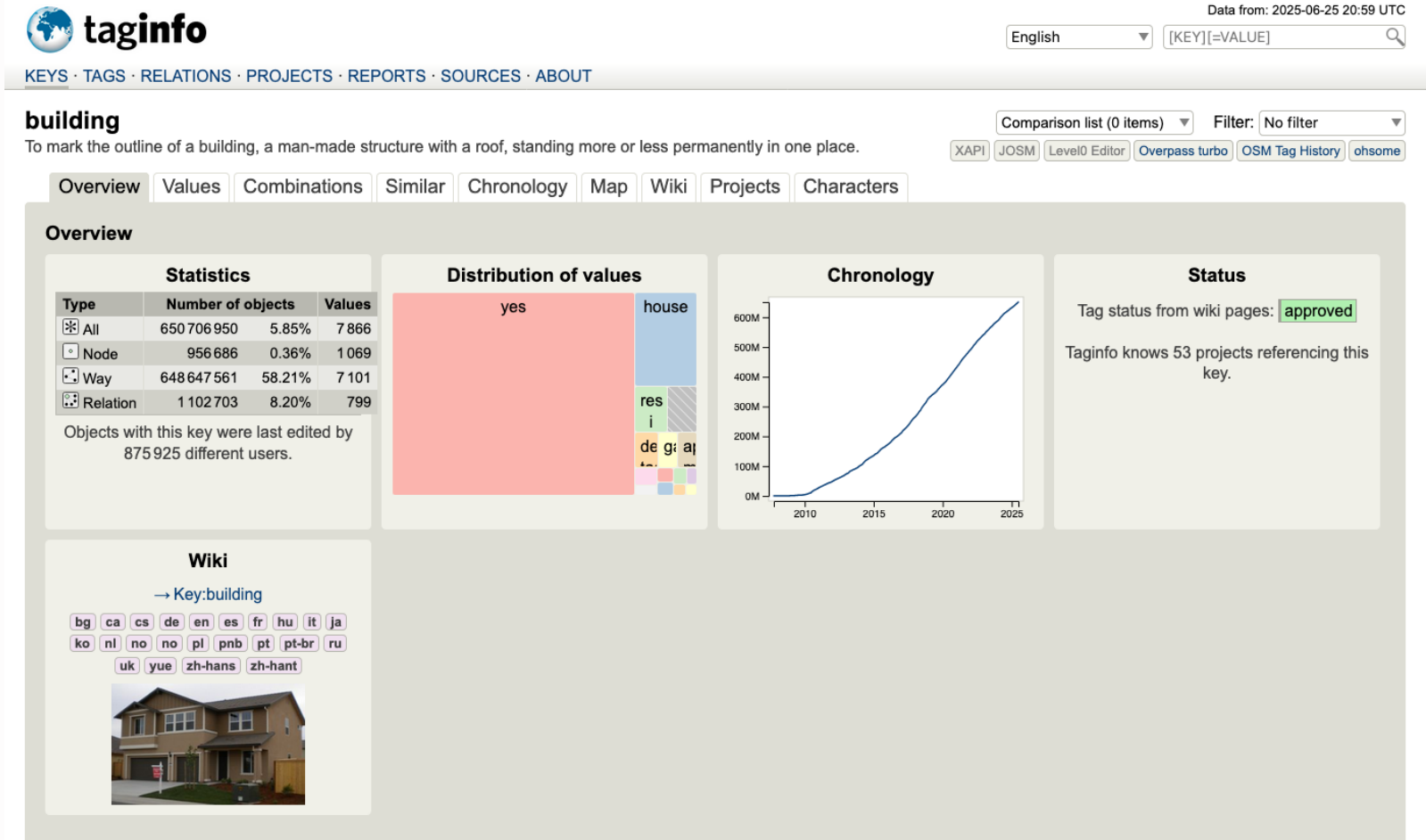
https://taginfo.openstreetmap.org/

Taginfo describes the actual uses of tags in the OSM (de facto).

- Most popular keys
- All possible values for a given key
- When possible, links to the wiki for more information

Linked from OSM wiki pages.







https://qgis.org



https://warwickcim.github.io/Inclusive_Cartographies_Workshop/



03
Lunch time!

We’ll be back in 45’!

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04 Cartography

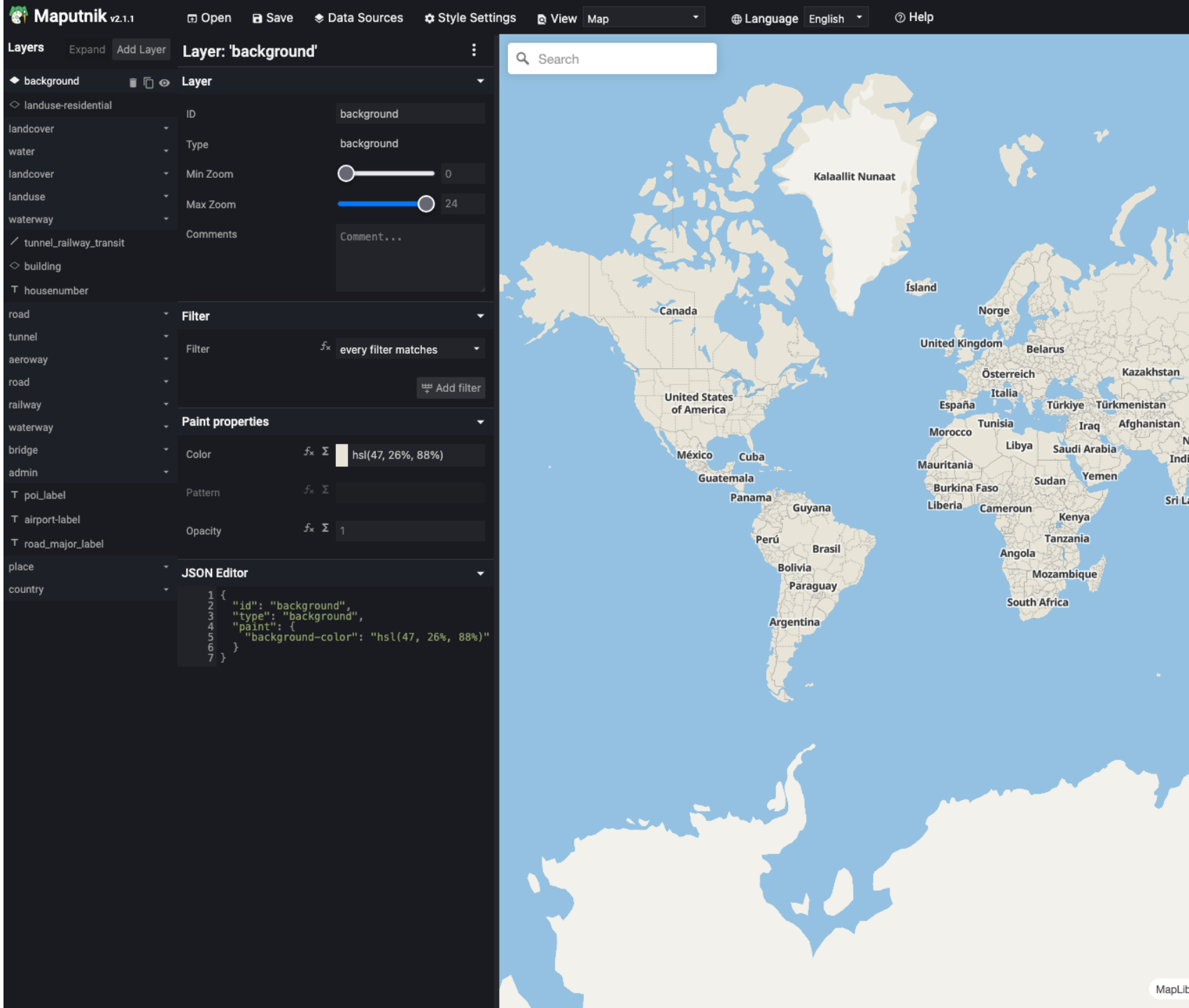
Designing a cartographic style for your map

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Maputnik

https://maplibre.org/maputnik

A free/libre online cartographic style editor for OpenStreetMap



Step 5: Turn your outline into a cartographic style

Use Maputnik to create a cartographic style (60’)

In groups or individually, try to convert your sketches and outline into a cartographic style

Once you’ve finished:

1. Export the JSON file
2. Share it with us:
    1. Save it this shared folder: https://drive.proton.me/urls/YDJ4TPQ4E8#GP5hIzDuHIgT
    2. Send it over email: carlos.camara@warwick.ac.uk / timothy.monteath@warwick.ac.uk

Share your work with us

Use Maputnik to create a cartographic style (60’)

QGIS:

1. Export to Geopackage
2. Put your files in this folder:



05 Discussion & Evaluation

We’d love to hear your thoughts!

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Discussion (30’)

Illustrative questions

1. Any thoughts on:
    1. Maps (and equity)
    2. OpenStreetMap
    3. What we did today
2. **Map features:**
    1. features that are available in OSM that you didn't know existed and are very useful/well covered,
    2. features that are not available, features that are not well covered,
3. **Design and design principles**
    1. Tips/ ideas…
    2. zoom scale that works well, and some other that doesn't
4. Is there anything you’ve done or learnt today you’d want to share with the participants?
5. How important would you say it is to have equitable maps?

Evaluation (5’)

Please, fill this short survey
https://shorturl.at/EhJQO

Thank you!



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